

```

# Number of students

n <- as.integer(
  readline("Enter number of students: ")
)

# Names

name <- character(n)

for(i in 1:n) {
  name[i] <- readline(
    paste("Enter name of student", i, ": ")
  )
}

# Scores

score <- numeric(n)

for(i in 1:n) {
  score[i] <- as.numeric(
    readline(
      paste("Enter score of", name[i], ": ")
    )
  )
}

# Attempts

attempts <- integer(n)

for(i in 1:n) {
  attempts[i] <- as.integer(
    readline(
      paste("Enter attempts of", name[i], ": ")
    )
  )
}

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    )
  )
}

# Qualification
qualify <- ifelse(score >= 50, "YES", "NO")

# Create data frame
exam_data <- data.frame(

  name,

  score,

  attempts,

  qualify
)

cat("\nOriginal Data Frame:\n")

print(exam_data)

# Extract score
cat("\nScores:\n")

print(exam_data$score)

# Extract first row
cat("\nFirst Student:\n")

print(exam_data[1, ])

# Extract qualified students
cat("\nQualified Students:\n")

print(exam_data[

  exam_data$qualify == "YES",

])

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# Add a new column

exam_data$grade <- ifelse(
  exam_data$score >= 75,
  "A",
  ifelse(exam_data$score >= 50, "B", "C")
)

cat("\nAfter Adding Grade Column:\n")

print(exam_data)

# Add a new row

new_name <- readline(
  "\nEnter new student name: "
)

new_score <- as.numeric(
  readline("Enter new student score: ")
)

new_attempts <- as.integer(
  readline("Enter new student attempts: ")
)

new_qualify <- ifelse(
  new_score >= 50,
  "YES",
  "NO"
)

new_grade <- ifelse(
  new_score >= 75,
  "A",

```

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    ifelse(new_score >= 50, "B", "C")
)
new_row <- data.frame(
  name = new_name,
  score = new_score,
  attempts = new_attempts,
  qualify = new_qualify,
  grade = new_grade
)
exam_data <- rbind(
  exam_data,
  new_row
)
cat("\nAfter Adding New Row:\n")
print(exam_data)
# Sort by score
exam_data <- exam_data[
  order(-exam_data$score),
]
cat("\nSorted by Score:\n")
print(exam_data)
# Save file
write.csv(
  exam_data,
  "exam_data.csv",
  row.names = FALSE

```

)

```
cat("\nData saved successfully as exam_data.csv\n")
```

The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains R code for adding a new row to a data frame, sorting by score, and saving to CSV.
- Environment:** Shows the `exam_data` object with 2 observations and 5 variables.
- Console:** Displays the execution output, including prompts for student information and the resulting data frame.

**Source Editor Code:**

```
86 }
87 grade = new_grade
88 exam_data <- rbind(
89   exam_data,
90   new_row
91 )
92 cat("\nAfter Adding New Row:\n")
93 print(exam_data)
94 # Sort by score
95 exam_data <- exam_data[
96   order(-exam_data$score),
97 ]
98 cat("\nSorted by Score:\n")
99 print(exam_data)
100 # Save file
101 write.csv(
102   exam_data,
103   "exam_data.csv",
104   row.names = FALSE
105 )
106 cat("\nData saved successfully as exam_data.csv\n")
107
```

**Environment:**

| Variable | Value                   |
|----------|-------------------------|
| attempts | int [1:2] 0 3           |
| i        | 2L                      |
| n        | 2L                      |
| name     | chr [1:2] "ravi" "rani" |
| qualify  | chr [1:2] "YES" "YES"   |
| score    | num [1:2] 99 85         |

**Console Output:**

```
> source("C:/Users/2021e/Desktop/EX 1.R")
Enter number of students: 2
Enter name of student 1 : ravi
Enter name of student 2 : rani
Enter score of ravi : 99
Enter score of rani : 85
Enter attempts of ravi : 0
Enter attempts of rani : 3

Original Data Frame:
  name score attempts qualify
1 ravi   99         0     YES
2 rani   85         3     YES

Scores:
[1] 99 85

First Student:
  name score attempts qualify
1 ravi   99         0     YES

Qualified Students:
  name score attempts qualify
1 ravi   99         0     YES
2 rani   85         3     YES

After Adding Grade Column:
  name score attempts qualify grade
1 ravi   99         0     YES    A
2 rani   85         3     YES    A

Enter new student name:
```